

Reference is made to the following documents:

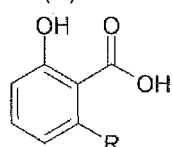
- D1 M Kim ET AL: "ABSTRACTS | Anacardic acid increases type I procollagen expression following ultraviolet irradiation via inhibition of DNA methylation in the COL1A2 promoter region", S117 Journal of Investigative Dermatology, 19 April 2014 (2014-04-19), XP055185170, Retrieved from the Internet: URL:<http://www.nature.com/jid/journal/v134/n1s/pdf/jid2014111a.pdf> [retrieved on 2015-04-22]
- D2 GHIZZONI M ET AL: "Improved inhibition of the histone acetyltransferase PCAF by an anacardic acid derivative", BIOORGANIC & MEDICINAL CHEMISTRY, PERGAMON, GB, vol. 18, no. 16, 15 August 2010 (2010-08-15), pages 5826-5834, XP027192824, ISSN: 0968-0896 [retrieved on 2010-07-03]
- D3 JP H6- 329 516 A (MIKIMOTO SEIYAKU KK) 29 November 1994 (1994-11-29)
- D4 DATABASE tkdl [Online] Indian patent office; 1915, "Dawa Barae Baras", XP002740788, Database accession no. NA4/1834
- D5 JP H10 36883 A (ASAHI CHEMICAL IND) 10 February 1998 (1998-02-10)
- D6 T. M. CALLAGHAN ET AL: "A review of ageing and an examination of clinical methods in the assessment of ageing skin. Part I: Cellular and molecular perspectives of skin ageing", INTERNATIONAL JOURNAL OF COSMETIC SCIENCE, vol. 30, no. 5, 1 October 2008 (2008-10-01), pages 313-322, XP055057484, ISSN: 0142-5463, DOI: 10.1111/j.1468-2494.2008.00454.x

Re Item I

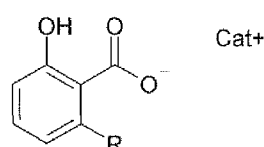
Basis of the report

1 The scope searched

- 1.1 The present application relates to the cosmetic use of at least one compound of formula (I), of at least one of its salts of formula (II) or of a mixture comprising at least one compound of formula (I) and at least one compound of formula (II):



(I)



(II)

as agent(s) intended to prevent and/or treat cutaneous signs of ageing.

- 1.2 In both formulae, R is claimed to independently represent a saturated, linear or branched, alkyl radical comprising from 12 to 20 carbon atoms.
- 1.3 However, the term "alkyl" is not defined in the claims, but based on the whole content of the description, it seems to depart from the standard IUPAC meaning. The description namely on p.3 exemplifies the compounds of formula (II) as those including unsaturations. Also according to the description, the term R is ambiguous. P. 7 defines it according to a first alternative form as not comprising an ethylenic unsaturation, but in a second alternative form, the R radical of the compounds according to the invention comprises at least one ethylenic unsaturation, preferably one, two or three ethylenic unsaturations, etc. Such alternatives are depicted also in the preparation on p.13.
- 1.4 Therefore the applicant was requested on 28.1.2016 to indicate if the searched subject matter is to be limited to the literal meaning of the claims (in this case only limited part of the description can be used for the support and the disclosure) or should the term R as being "saturated" be disregarded and alkyl interpreted as saturated or unsaturated in line with the description. The applicant desired with the letter of 10.2.2016 the search to be limited to the literal meaning of the claims

Re Item II

Priority

2 Limitation compared to the priority application

- 2.1 The priority application related to the use of compounds of formula (I) or (II) which were much broader than the presently claimed subject matter. The limitation to only saturated compounds is a selection within the general teaching of the priority application and goes against the general teaching of the priority application. E.g. in claim 6 or on p.9-10 of the priority application a selection of the minority of the depicted compound is necessary to arrive at the presently claimed compounds.

Re Item V

Reasoned statement with regard to novelty, inventive step or industrial applicability; citations and explanations supporting such statement

3 Novelty

- 3.1 **D1** discloses that anacardic acid (AA) has a significant effect on the increase in the expression of procollagen, which counteracts "photoaging" of the skin. Inherently the anacardic acid has been used in a composition.
- 3.1.1 Reference is made to **D2**, which is a scientific article in the field of histone acetylation (see the third line in the abstract of this document) which is exactly the field of **D1** (see the third line of **D1**). **D2** depicts on Fig 1 and on Scheme 1 exactly the 6-pentadecylsalicylic acid as the anacardic acid. In this regard it is noted that both documents use the same abbreviation (AA) for this compound.
- 3.1.2 This authority is not in the position to retrieve the full content of the presentation abstracted by the document **D1**, however assumes that the disclosure AA to counteract the "photoaging" of the skin corresponds to the the subject matter of the present claims 1,2,9, 12 and 14.
- 3.2 This application explains that "the signs of skin aging" include a lack of uniformity of the complexion. The acids of formula (I) (as anacardic acid) are known for bleaching of the skin, which-is considered as treatment of the lack of uniformity of the complexion. **D3** teaches cosmetic composition having

between 0.05% (eg. 11) and 5% (ex. 6) by weight of such acids to combat patches of skin pigmentation (see paragraph 2 and 26), particularly for skin whitening (see paragraph 4 and 27). In particular this document (in the examples on p.3-6) teaches 6-pentadecylsalicylic acid which is the compound A' of present claims 9. In example on. p.3 this document teaches the combination of 6-pentadecylsalicylic acid with potassium hydroxide. Such combination will result in formation of the salt of formula (II), which falls under present claim 3.

3.2.1 Therefore, claims 1,2,3, 9,12-14 lack novelty vis-a-vis this document.

3.3 Similarly, the subject matter of at least claim 1 is not new vis-à-vis **D4**, which provides use of anacardium (which contains anacardic acid) against vitiligo (which is also a lack of uniformity of the complexion).

4 Inventive step

4.1 The application relates to the use of the compound of the acids of formula (I) or their salts (of formula (II)) for preventing and / or treating cutaneous signs of aging.

4.2 The specific compound of examples (Compound A 'of this application), recited in claim 9 is one of the compounds used in **D3** to combat uneven skin pigmentation (see skin stains mentioned in para [0002]. Hence even the preferred compound of the present application has been used against the signs of skin aging (skin pigmentation stains is the change to the external appearance of the skin which is skin aging according to the present application). The same compound is the (AA) of **D1** disclosed as effective against skin photoaging.

5 In view of the lack of novelty, the following must be considered for the assessment of inventive step if the applicant is able to restore novelty.

5.1 **D3** and **D1** both relate to the present compounds and the field of use:

5.1.1 Structurally, compared to **D3**, the subject matter of the present application differs by either converting the 6-pentadecylsalicylic acid into salts with amino acids (claims 4 to 8) or using them against wrinkles, fine lines, or withered, flabby and/or thinned skin (claims 10,11).

- 5.1.2 Mechanistically, **D1** discloses that anacardic acid has a significant effect on the increase in the expression of procollagen to combat "photoaging" of the skin. The subject matter of the present application differs by converting this (or similar) compounds into salts with amino acids (claims 4 to 8) or using them against wrinkles, fine lines, or withered, flabby and/or thinned skin (claims 10,11).
- 5.1.3 For the sake of the argument, due to the same scientific mechanism underlying the present example 2 (procollagen synthesis), this document is taken as the closest prior art.
- 5.1.4 The present application in generally tested only the compound falling under the salts of formula (I) (the compound F) in example 2 for anti-aging, but the compounds A' and F (free acid of formula (I) and its salt) for a very specific use to combat withered skin in example 3. Therefore the objective technical problems which are solved by the presently claimed subject matter (in as far as novel) may be seen in the provision of the formulation of the anacardic acid or the specific use thereof.
- 5.1.5 In one case, the solution is proposed by the formulation as the amino acid salt and in the other case, by the specific use.
- 5.2 **D1** on its own does not disclose how the anacardic acid should be formulated. However a skilled person working in the field of cosmetics is aware of **D5**. This document discloses (see para [0012]) that free anacardic acid is irritating and causes skin rash. To solve this problem, this documents proposes to convert it into salts (see para [0014]-[0015]) and proposes i.a. basic amino acids ((see para [0019]-[0020]), such as lysine (ex.2).
- 5.2.1 Consequently starting from the mechanicist study elaborated in **D1**, a skilled person wanting to formulate the tested compound for cosmetic purposes would be motivated by **D5** to avoid skin rash and would employ the formulation strategy (salt formation) proposed by **D5**. The skilled person would have primarily used lysine as exemplified by **D5** and arrive at the present specific compound F of claim 8. The solution proposed in claims 4-8 is this obvious.
- 5.3 For the specific uses of claims 10 and 11, a skilled person is aware of the review article **D6**, which is considered as a part of a general knowledge in the filed of collagen synthesis (the field of **D1**). This review article explains the similarities between photoaged and chronologically aged skin and indicates (see abstract) such skin as thin, flattened, dry and unblemished as well as

loosing architectural regularity. These are considered synonyms to herein claimed signs of aging recited in claims 10 and 11 and the skilled person would expect the compounds active on procollagen synthesis to alleviate these signs.

Re Item VIII

Certain observations on the international application

6 Clarity of the terminology

- 6.1 The term "cutaneous signs of aging" has been interpreted broadly enough, because according to the present application (p.5-6) these signs include any changes to the external appearance of the skin including also changes of the complexion.
- 6.2 The term alkyl used in the claims is not compatible with the whole disclosure of the application. The alkyl is defined as saturated, but the description exemplifies (see p.9) as the very particularly preferred form of the invention the compounds in which alkyl is not saturated.